S1 = ABCDGH
S2 = AEDFHR

	A	E	D	F	H	R
A	③	-1	-1	-1	-1	-1
B	-1	2	2	1	1	0
C	-1	2	2	1	1	0
D	-1	2	2	1	1	0
G	-1	1	1	1	1	0
H	-1	1	1	1	1	0

TC = O(nm).

x = nm
y = 1

Top-Down.

TC = Find the no. of unique subproblems solved → x
Time taken to solve a single subproblem → y.

TC = O(xy).

	A	E	D	F	H	R
A	1	1	1	1	1	1
B	1	1	1	1	1	1
C	1	1	1	1	1	1
D	1	1	2	2	2	2
G	1	1	2	②	2	2
H	1	1	2	2	3	③

dp[i][j] → LCS of s1[0 ... i] & s2[0 ... j].

dp[i][j] = 1 + dp[i-1][j-1] if chars are same
max (dp[i-1][j], dp[i][j-1]) otherwise.

	0	A 1	E 2	D	F	H	R
0	0	0	0	0	0	0	0
A 1	0						
B 2	0						
C 3	0						
D	0						
G	0						
H	0						

dp[i][j] → LCS of s1[0 --- (i-1)] &
s2[0 --- (j-1)].

---

vals: [60 100 120]
wts: [10 20 30]
W = 50
o/p = 220.

vals: [60 100 120]
wts: [10 20 50]
W = 50
o/p = 160.

n_items.

$\frac{2}{0}\ \frac{2}{}\ ------\ \frac{2}{n-1}$

2×2×2× ---- = 2^n combinations possible.

0 1 2

[]
0
1
2
0 1
0 2
1 2
0 1 2

↓
vals: [60 100 120]
wts: [10 20 30]
W = 50.

exclude
vals: [100, 120]
wts: [20, 30]
W = 50

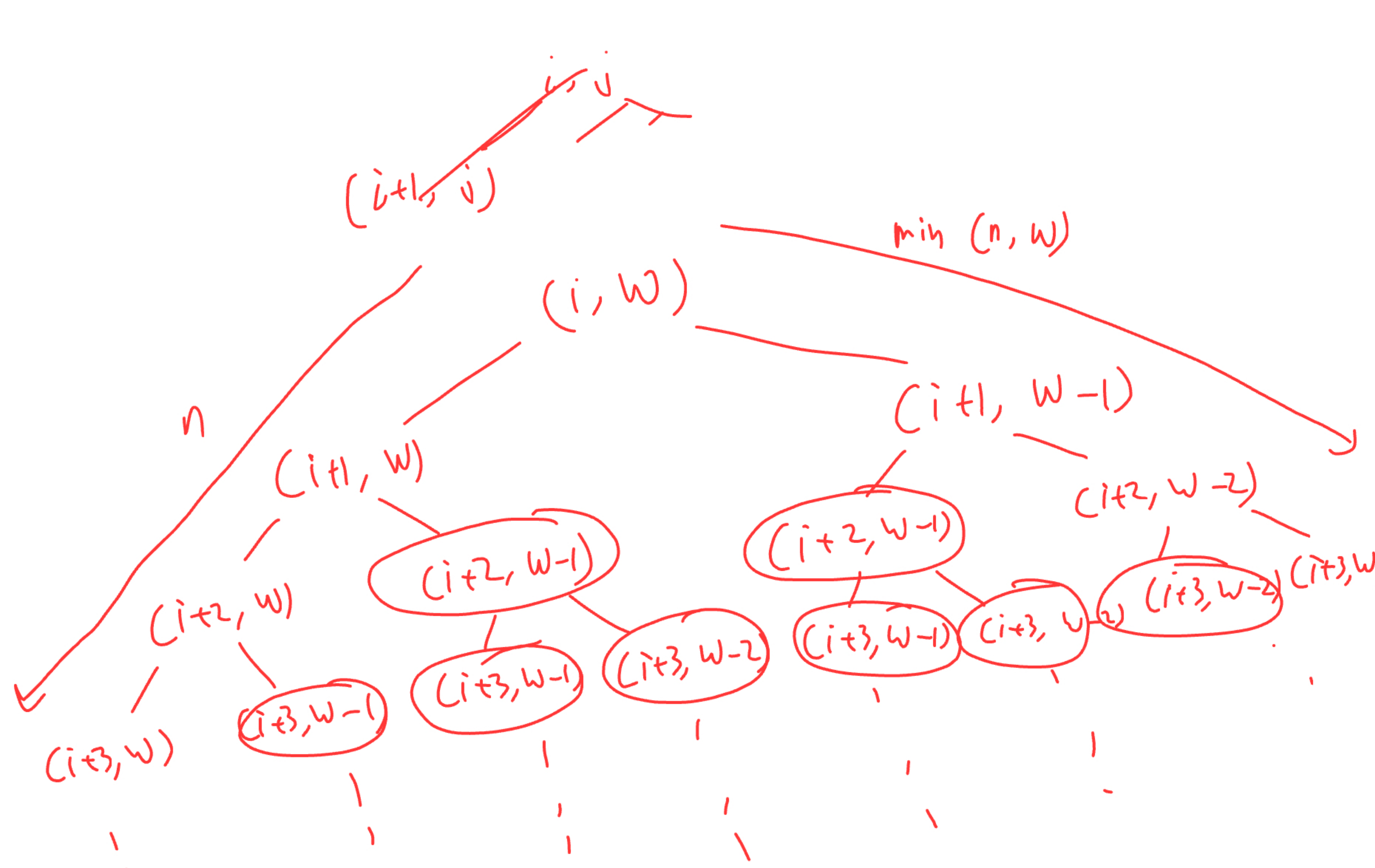
include
vals: [100, 120]
wts: [20, 30]
W = 40.
+ 60.

vals: [100]
wts: [50]
W = 20.

0+100 = (100)

vals: []
wts: []
W = -30.

maxValue(i, j) = max (maxValue (i+1, j),
maxValue (i+1, j - wts[i]))



(i, j)
(i+1, j)
(i, W)
min (n, W)
n
(i+1, W)
(i+1, W-1)
(i+2, W)
(i+2, W-1)
(i+2, W-2)
(i+3, W)
(i+3, W-1)
(i+3, W-2)
(i+3, W-1)
(i+3, W-2)
(i+3, W-3)
(i+3, W-3)

H = max (n, min(n, W)).
= n

TC = O(2^n).
AS: O(n).

A = Anshuman
B = Antihuman
↓
(nshuman, ntihuman)
↓
(shuman, tihuman)

ED(i, j) = ED(i+1, j+1)
if A[i] == B[j].

Insert
(shuman, ihuman)
[i, j+1].

Delete
(human, tihuman)
[i+1, j]

Replace
(human, ihuman).
[i+1, j+1].

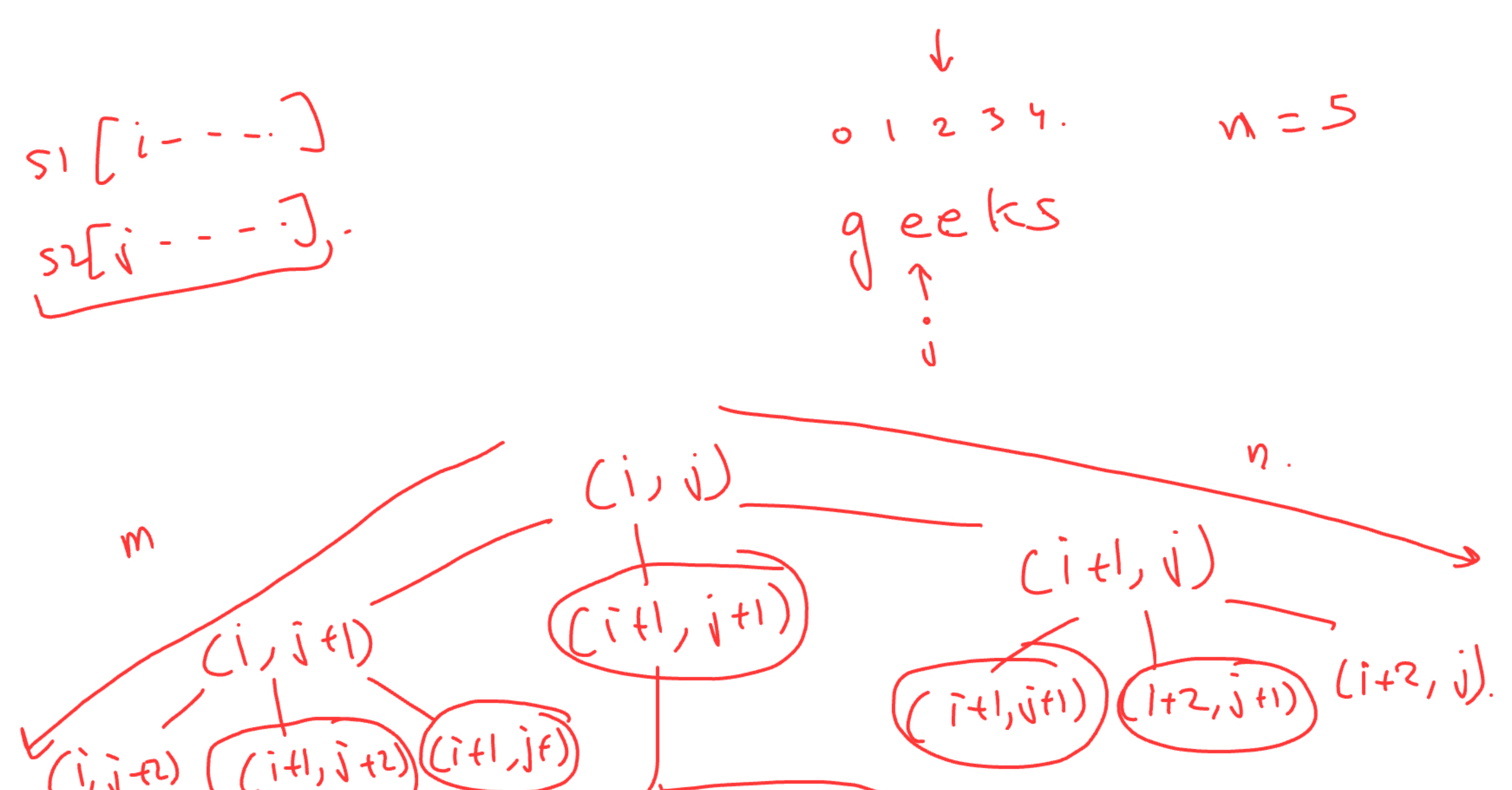
Min.

ED(i, j) = ED(i+1, j+1) if A[i] == B[j].
1 + min [ED(i, j+1), ED(i+1, j), ED(i+1, j+1)] otherwise.

↓
0 1 2 3 4.
geeks
↑
i

n = 5

s1[i - - -]
s2[j - - -].



m
n.
(i, j)
(i, j+1)
(i+1, j+1)
(i+1, j)
(i, j+2)
(i+1, j+2)
(i+1, j+1)
(i+1, j+1)
(i+2, j+1)
(i+2, j)
(i+1, j+2)
(i+2, j+2)
(i+2, j+1)

H = max(n, m).
TC = $O(3^{\max(n,m)})$.
AS = max (n, m).

---

coins: [2, 3, 5, 6]
(i, value).
value = 10.

exc.
[3, 5, 6].
10
(i+1, val).

inc
+1
[2, 3, 5, 6].
value = 8
(i, val - coins[i]).

~~10~~
4
1

coins: [5, 6, 8, 9].
value = 11
-1
2
-1
1
[6, 8, 9]
11
[5, 6, 8, 9].
val = 6